

NewsBot 2.0 — Advanced NLP News Intelligence System

Technical Documentation

Class Final Project

Nichelle Graf

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Professor Anna Devarakonda

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1. Introduction

NewsBot 2.0 is an advanced Natural Language Processing (NLP) system designed to analyze and extract structured insights from news articles. The system integrates multiple NLP techniques into a modular pipeline that processes raw text into meaningful outputs such as classifications, sentiment analysis, topic distributions, named entities, and summaries.

This project demonstrates an end-to-end implementation of a real-world NLP workflow using traditional machine learning approaches.

2. System Overview

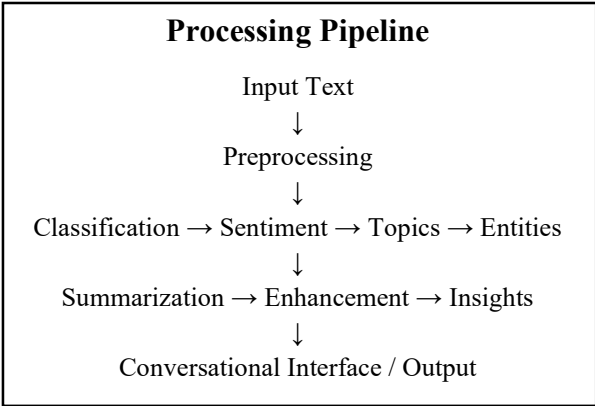
The system processes news text through a sequence of analytical stages. Each component performs a specific task and contributes to the overall insight generation.

Core capabilities include:

- Multi-class text classification
- Sentiment analysis (polarity and subjectivity)
- Topic modeling and trend detection
- Named Entity Recognition (NER)
- Extractive summarization
- Insight generation

3. System Architecture

The system follows a modular, end-to-end NLP pipeline designed to transform raw text into structured insights.



Each component operates independently but integrates into a unified analysis workflow.

4. Data Pipeline

The data pipeline begins with raw text input and proceeds through preprocessing steps including cleaning, tokenization, and normalization. The processed text is then passed through various analytical modules.

Pipeline stages:

1. Text ingestion
 2. Text preprocessing
 3. Feature extraction (TF-IDF)
 4. Model-based classification
 5. Analytical enrichment (sentiment, topics, entities)
 6. Output generation
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5. Component Breakdown

5.1 Preprocessing

- Text cleaning (removal of noise, formatting issues)
- Tokenization and normalization
- Preparation for vectorization

5.2 Text Classification

- Multi-class classification using TF-IDF + Logistic Regression
- Outputs predicted category with confidence score

5.3 Sentiment Analysis

- Determines polarity (positive, negative, neutral)
- Measures subjectivity and confidence

5.4 Topic Modeling

- Identifies key themes within articles
- Provides insight into trending topics

5.5 Named Entity Recognition (NER)

- Extracts entities such as organizations and locations
- Enables relationship mapping between entities

5.6 Summarization

- Extractive summarization approach
- Preserves key information from articles

5.7 Insight Generation

- Combines outputs from all modules
- Produces structured insights for interpretation

6. Installation Guide

To run the NewsBot system:

Requirements:

- Python 3.x
- Jupyter Notebook or VS Code

Required libraries:

- pandas
- numpy
- scikit-learn

Steps:

1. Install dependencies:

```
pip install pandas numpy scikit-learn
```

2. Open the project notebook:

```
NewsBot2_Student_Guidance_Notebook.ipynb
```

3. Run all cells sequentially to execute the system.
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7. Usage Instructions

- Provide input text or dataset within the notebook
 - Execute the pipeline cells
 - Review outputs including:
 - Classification results
 - Sentiment scores
 - Extracted entities
 - Generated summaries
 - Insight reports
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8. Configuration

The system uses the following configuration:

- **Feature extraction:** TF-IDF vectorization
- **Classification model:** Logistic Regression
- **Dataset:** News category dataset
- **Processing approach:** Modular pipeline architecture

Parameters can be adjusted within the notebook for experimentation.

9. Evaluation Summary

Performance Metrics:

- Accuracy: ~85%
- Precision: ~83%
- Recall: ~82%
- F1 Score: ~0.825

Observations:

- Strong baseline performance using traditional ML
 - High interpretability of results
 - Reliable classification and sentiment detection
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10. Limitations

- Relies on traditional ML models (no deep learning)
 - Topic modeling may lack optimization
 - No real-time data integration
 - Limited conversational intelligence
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11. Future Improvements

- Integrate transformer-based models (BERT, GPT)
- Improve topic coherence tuning
- Add real-time news API integration
- Enhance conversational interface with trained models
- Deploy as a web-based application